GLOBAL EDITION

MICROECONOMICS

14th Edition

Michael Parkin



3 DEMAND AND SUPPLY

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After studying this chapter, you will be able to:

- Describe a competitive market and think about a price as an opportunity cost
- Explain the influences on demand
- Explain the influences on supply
- Explain how demand and supply determine prices and quantities bought and sold
- Use the demand and supply model to make predictions about changes in prices and quantities

↳ how to predict D vs S based on current mood

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Markets and Prices

location where buyers + sellers interact

A *market* is any arrangement that enables buyers and sellers to get information and do business with each other.

continuous demand leads to a boost in prices

A **competitive market** is a market that has many buyers and many sellers so no single buyer or seller can influence the price.

how buyers & sellers influence

The **money price** of a good is the amount of money needed to buy it. e.g. 10 SAR, 15 SAR, etc - helps determine whether you will buy the product or not

The **relative price** of a good—the ratio of its money price to the money price of the next best alternative good—is its *opportunity cost*.

substitute item that almost gives you the same satisfaction

price changes affect budget driving low-income ppl to sacrifice

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Demand

If you demand something, then you

1. Want it,
2. Can afford it, and
3. Have made a definite plan to buy it.

Wants are the unlimited desires or wishes people have for goods and services. Demand reflects a decision about which wants to satisfy.

The **quantity demanded** of a good or service is the amount that consumers plan to buy during a particular time period, and at a particular price.

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sellers → producers

Demand

The Law of Demand

→ focus is the price of the product

The **law of demand** states:

constant

Other things remaining the same, the higher the price of a good, the smaller is the quantity demanded; and …

different prices correspond to change in demand

the lower the price of a good, the larger is the quantity demanded.

P↑ Qd↓ , P↓ Qd↑ → establish negative relationship

Why does a change in the price change the quantity demanded? Two reasons:

↑ $100 → 500
$200 → 250 ↓

1. Substitution effect
   - look for alternative items at a lower price that offers the same satisfaction
2. Income effect

} factors that explain this law

/budget

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Demand

Substitution Effect

When the relative price (opportunity cost) of a good or service rises, people seek substitutes for it, so the quantity demanded of the good or service decreases.

Income Effect

When the price of a good or service rises relative to income, people cannot afford all the things they previously bought, so the quantity demanded of the good or service decreases.

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Demand

Demand Curve and Demand Schedule

The term **demand** refers to the entire relationship between the price of the good and quantity demanded of the good.

A **demand curve** shows the relationship between the quantity demanded of a good and its price when all other influences on consumers' planned purchases remain the same.

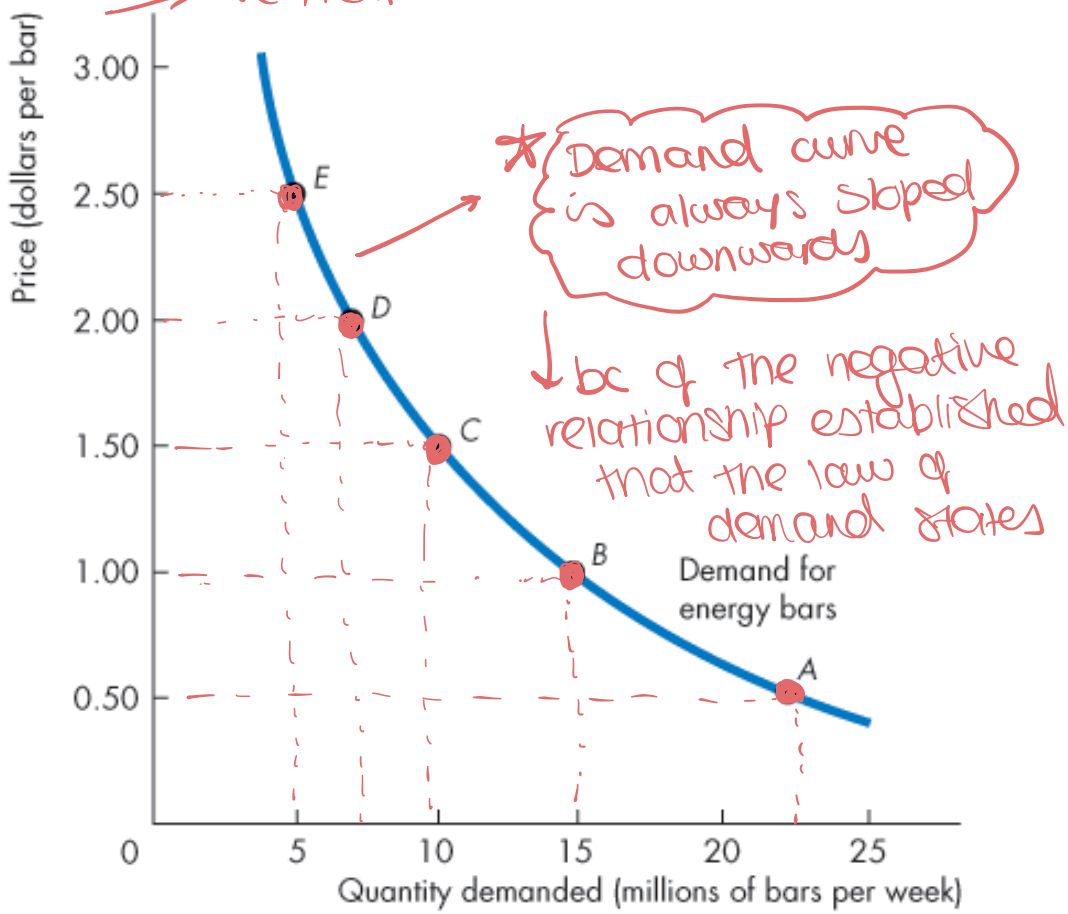


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Demand

Figure 3.1 shows a demand curve for energy bars.

	Price (dollars per bar)	Quantity demanded (millions of bars per week)
A	0.50	22
B	1.00	15
C	1.50	10
D	2.00	7
E	2.50	5



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Demand

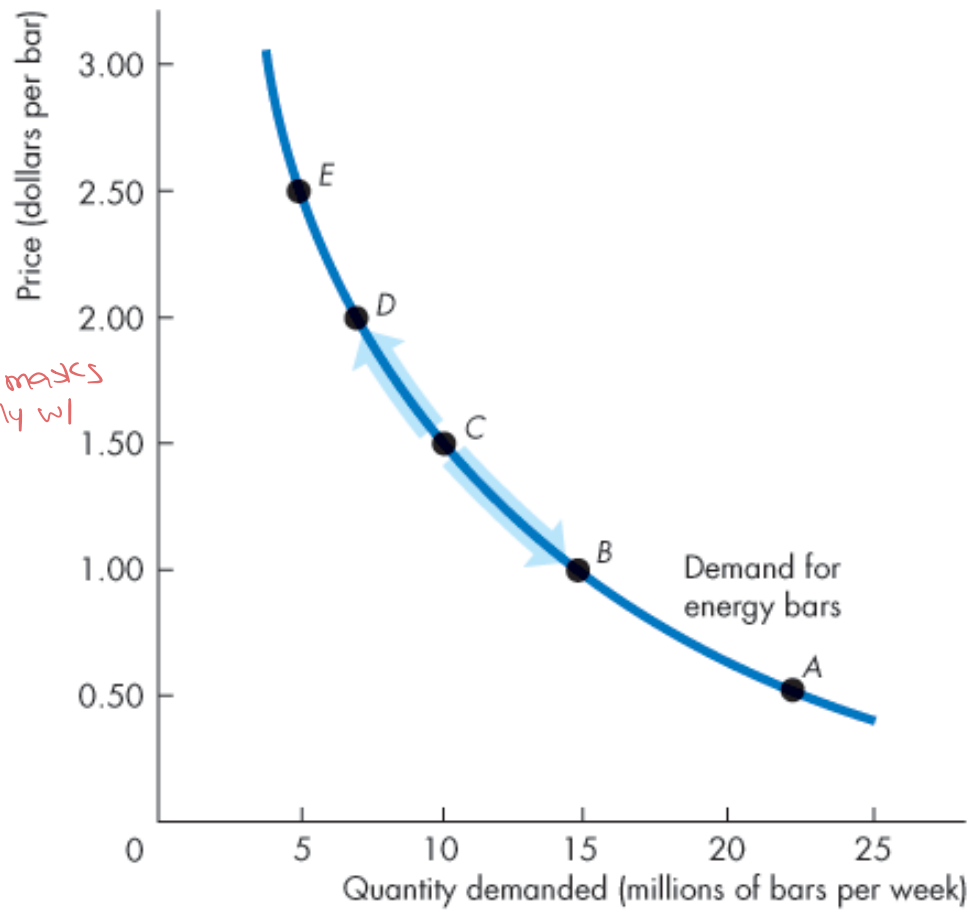
- movement along the curve indicates changes in price
- shift indicates changes in demand

will be covered

A rise in the price, other things remaining the same, decreases the quantity demanded and brings a movement up along the demand curve.

other factors e.g choices, style, fashion, law e.g wearing masks to comply w/ laws

A fall in the price, other things remaining the same, increases the quantity demanded and brings a movement down along the demand curve.



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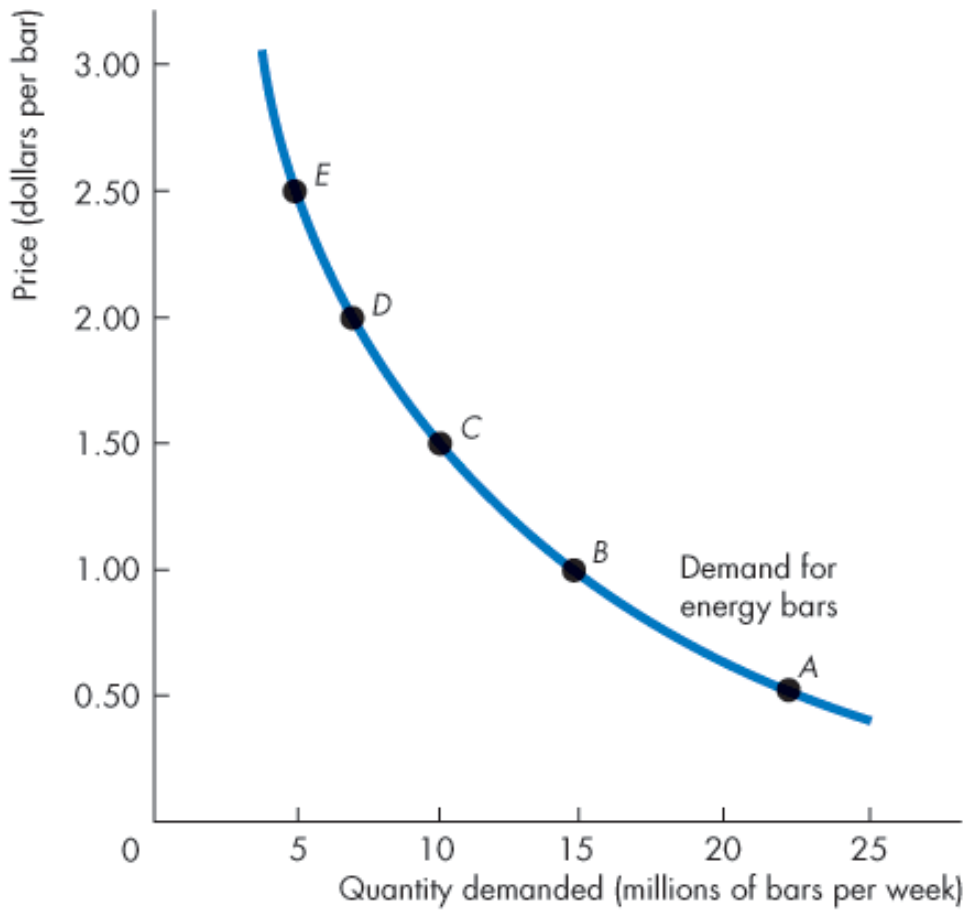
Demand

Willingness and Ability to Pay

A demand curve is also a *willingness-and-ability-to-pay* curve.

As the quantity available decreases, the higher is the price that someone is willing to pay for another unit.

Willingness to pay measures *marginal benefit*.



each additional value consumed tends to lose its value overtime

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Demand

A Change in Demand

When some influence on buying plans other than the price of the good changes, there is a **change in demand** for that good.

The quantity of the good that people plan to buy changes at each and every price, so there is a new demand curve.

When demand *increases*, the demand curve shifts *rightward*.

When demand *decreases*, the demand curve shifts *leftward*.

→ income expectations determine demand

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weather destroys crops, so coffee beans will become expensive → so demand

complementary items: mobile & charger, printer & an ink cartridge, car & gasoline

when the model of a mobile becomes outdated its corresponding accessories decrease in demand

Demand

in the current market increases → curve shifts to the right (if you expect a higher price)

price of one market will affect the demand of another
e.g. demand of NADIC increases, because Amarai is expensive

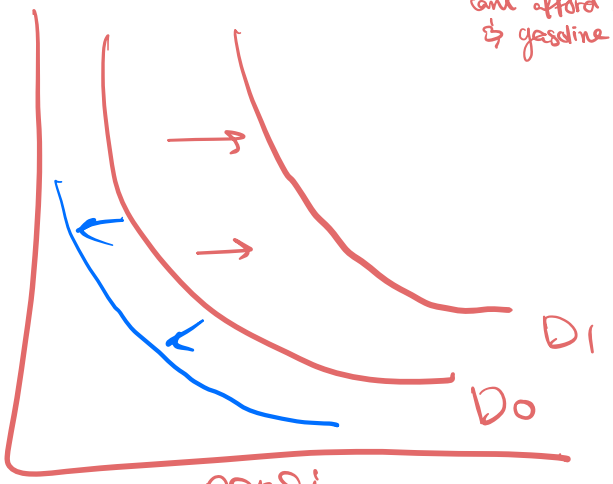
Six main factors that change demand are:

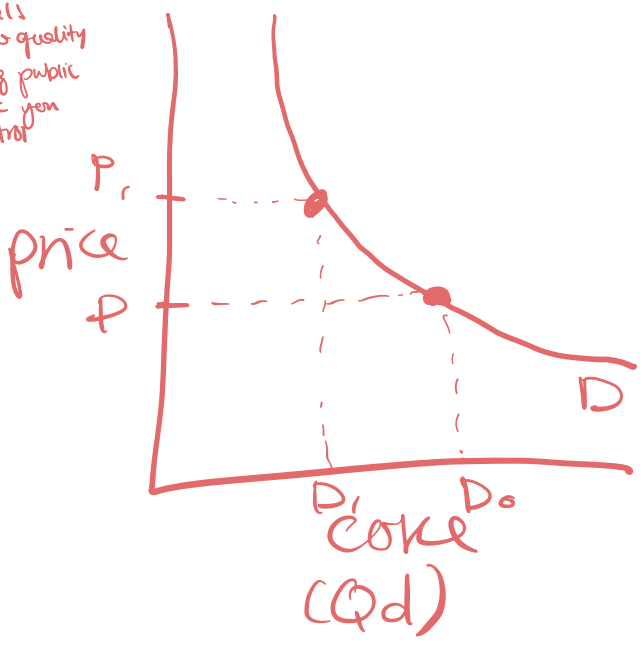
→ six factors / shifts determinants

- The prices of related goods
  - → complementary & substitutes
- Expected future prices
  - → discounts on occasional events in the near future
  - demand in the current market decreases as a result
  - → curve shifts to the left
  - ANALYSIS IS DONE ON THE CURRENT MARKET
  - Coke market example
- Income
  - as income increases, demand increases
  - two types of goods: normal & inferior goods - normal goods has a positive relationship with income levels
  - has a negative relationship relative to income levels
  - lower-income individuals resort to cheap-low quality products e.g. using public transportation bc you can't afford petrol & gasoline
- Expected future income and credit
- Population
  - → establishes a positive relationship
- Preferences
  - → subjective opinions & likings shape demand

curve will shift to the left, if Pepsi decide to lower its prices - public gets attracted to this

Price





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market, decreasing the demand that its competitor has

Demand

Prices of Related Goods

A **substitute** is a good that can be used in place of another good. → good that can be used alternatively of another

A **complement** is a good that is used in conjunction with another good.

When the price of a substitute for an energy bar rises or when the price of a complement of an energy bar falls, the demand for energy bars increases.

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Demand

Expected Future Prices

If the price of a good is expected to rise in the future, current demand for the good increases and the demand curve shifts rightward.

Income

When income increases, consumers buy more of *most* goods and the demand curve shifts rightward.

A **normal good** is one for which demand increases as income increases.

An **inferior good** is a good for which demand decreases as income increases.

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Demand

Expected Future Income and Credit

When income is expected to increase in the future or when credit is easy to obtain, the demand might increase now.

Population

The larger the population, the greater is the demand for all goods.

Preferences

People with the same income have different demands if they have different preferences.

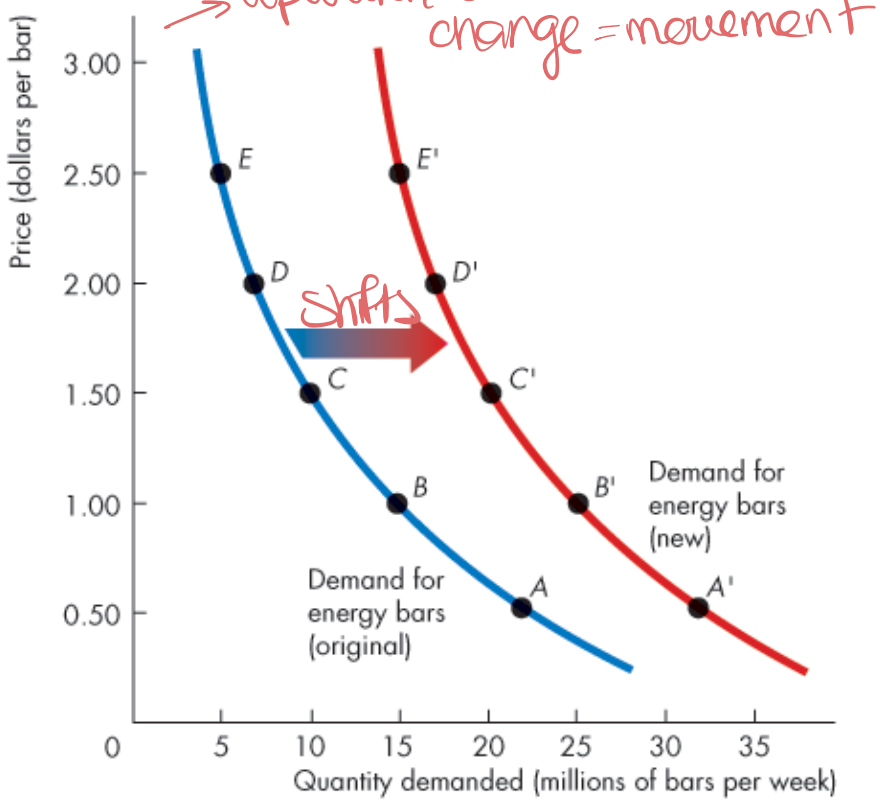
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Demand

Figure 3.2 shows an increase in demand.

An increase in income increases the demand for energy bars and shifts the demand curve rightward.

	Original demand schedule Original income			New demand schedule New higher income	
	Price (dollars per bar)	Quantity demanded (millions of bars per week)		Price (dollars per bar)	Quantity demanded (millions of bars per week)
A	0.50	22	A'	0.50	32
B	1.00	15	B'	1.00	25
C	1.50	10	C'	1.50	20
D	2.00	7	D'	2.00	17
E	2.50	5	E'	2.50	15

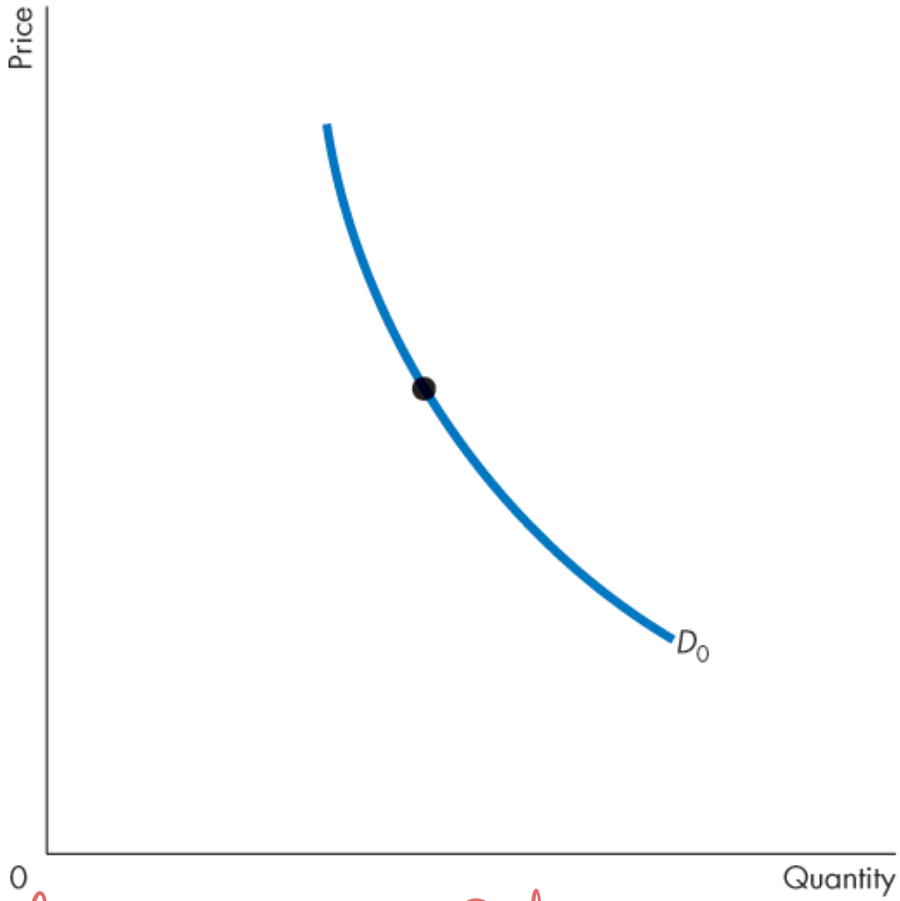


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Demand

A Change in the Quantity Demanded Versus a Change in Demand

Figure 3.3 illustrates the distinction between a change in demand and a change in the quantity demanded.



Δd

ΔQd

shift in the curve → demands based on 6 factors

movement in the curve → Quantity demand based on price changes

Qd

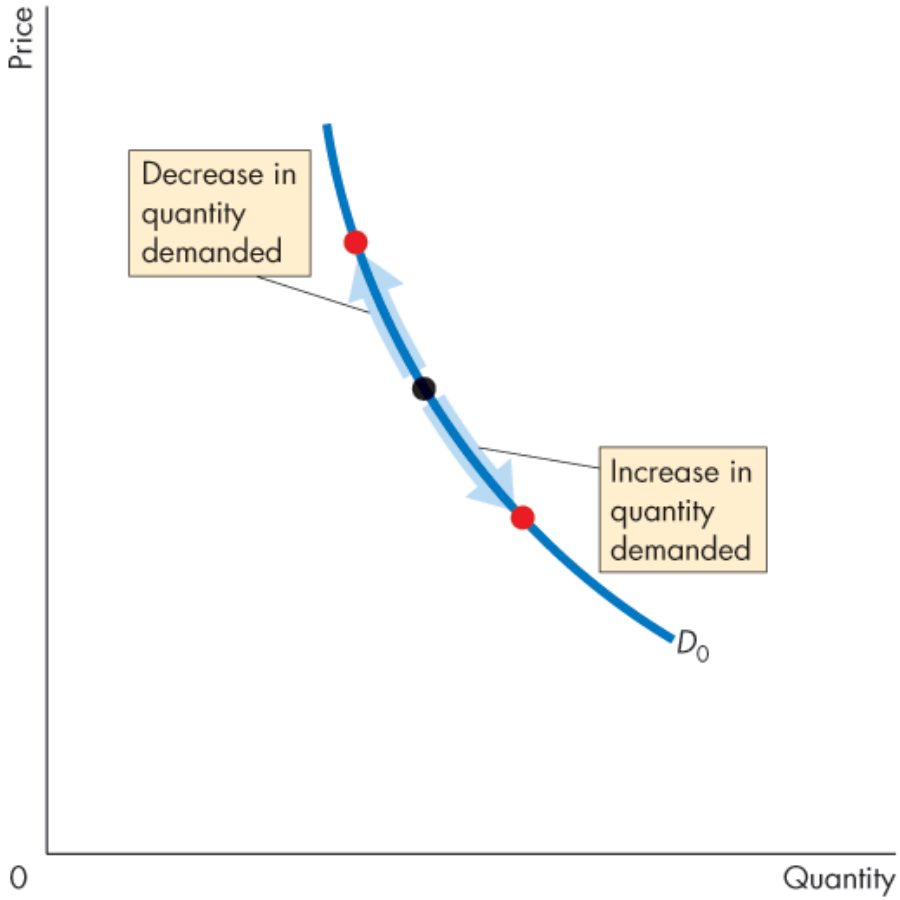
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★ income declines, what happens to small cars → demand increase
price of gasoline → ambigious → demand is ambigious

Demand

Movement Along the Demand Curve

When the price of the good changes and other things remain the same, the quantity demanded changes and there is a movement along the demand curve.

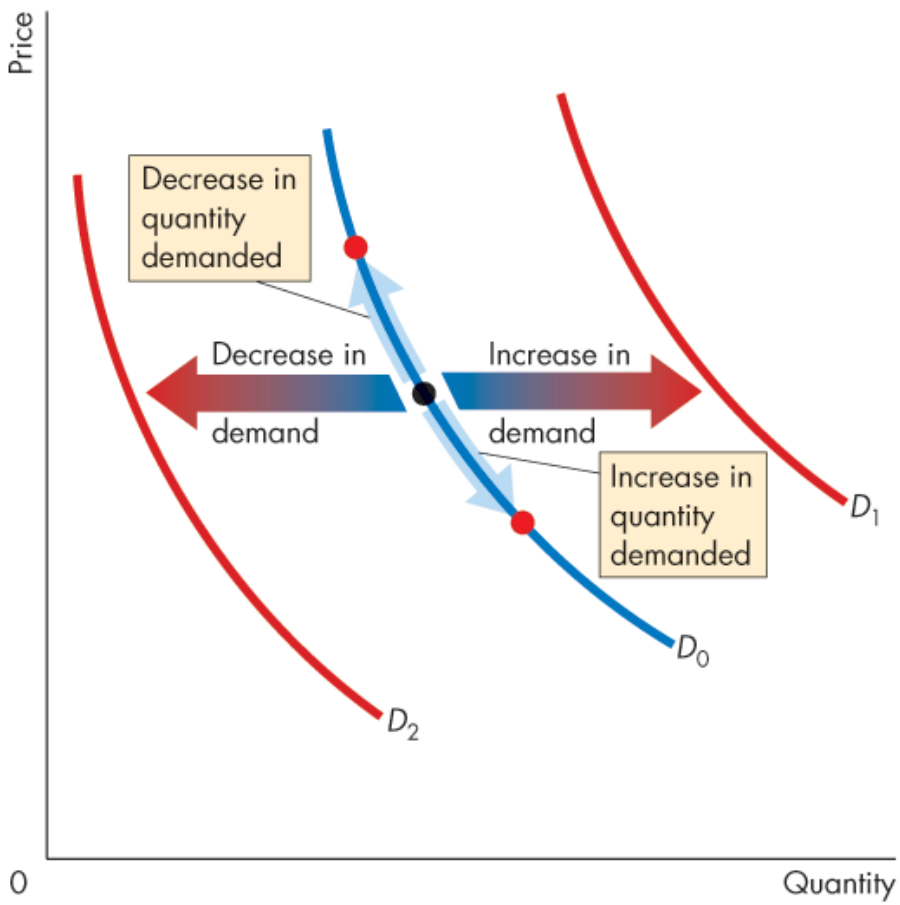


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Demand

A Shift of the Demand Curve

If the price remains the same but one of the other influences on buyers' plans changes, demand changes and the demand curve shifts.



→ cost of production is factor causing the prices to increase unless there is shortage in market (less production, more demand)

(sellers) → seek high prices to gain maximize profit

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Supply

If a firm supplies a good or service, then the firm

1. Has the resources and the technology to produce it,
2. Can profit from producing it, and
3. Has made a definite plan to produce and sell it.

Resources and ***technology*** determine what it is possible to produce. Supply reflects a decision about which technologically feasible items to produce.

The **quantity supplied** of a good or service is the amount that producers plan to sell during a given time period at a particular price.

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Supply

The Law of Supply

The **law of supply** states:

Other things remaining the same, the higher the price of a good, the greater is the quantity supplied; and

the lower the price of a good, the smaller is the quantity supplied.

The law of supply results from the general tendency for the marginal cost of producing a good or service to increase as the quantity produced increases (Chapter *2,* page 35).

Producers are willing to supply a good only if they can at least cover their marginal cost of production.

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Supply

Supply Curve and Supply Schedule

The term **supply** refers to the entire relationship between the quantity supplied and the price of a good.

The **supply curve** shows the relationship between the quantity supplied of a good and its price when all other influences on producers' planned sales remain the same.

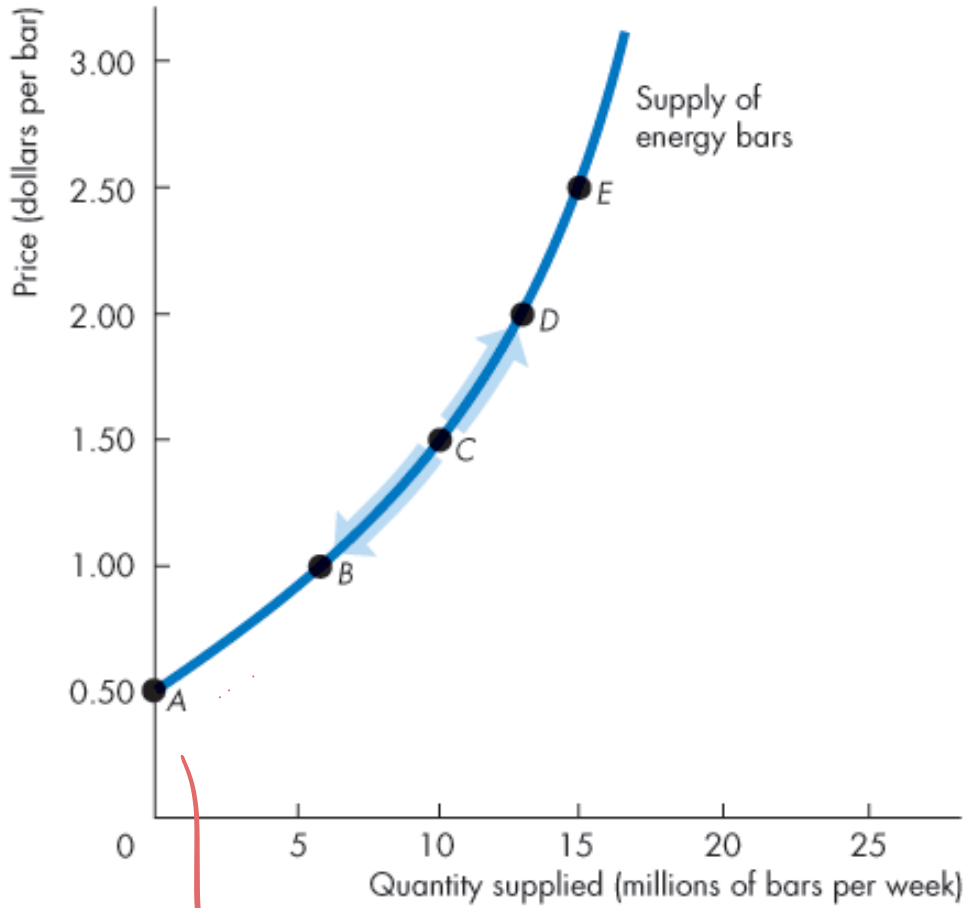
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Supply

Figure 3.4 shows a supply curve of energy bars.

	Price (dollars per bar)	Quantity supplied (millions of bars per week)
A	0.50	0
B	1.00	6
C	1.50	10
D	2.00	13
E	2.50	15

A rise in the price, other things remaining the same, brings an increase in the quantity supplied.



upward slope due to positive relationship

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Supply

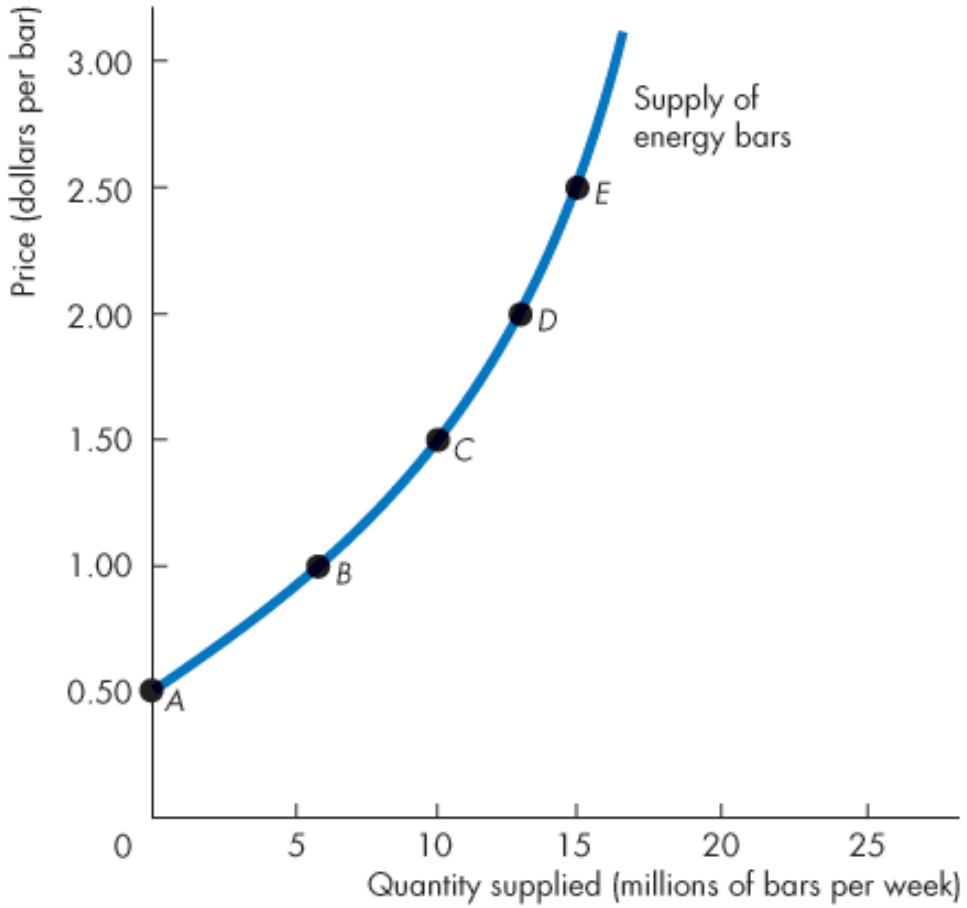
Minimum Supply Price

A supply curve is also a *minimum-supply-price* curve.

As the quantity produced increases, marginal cost increases.

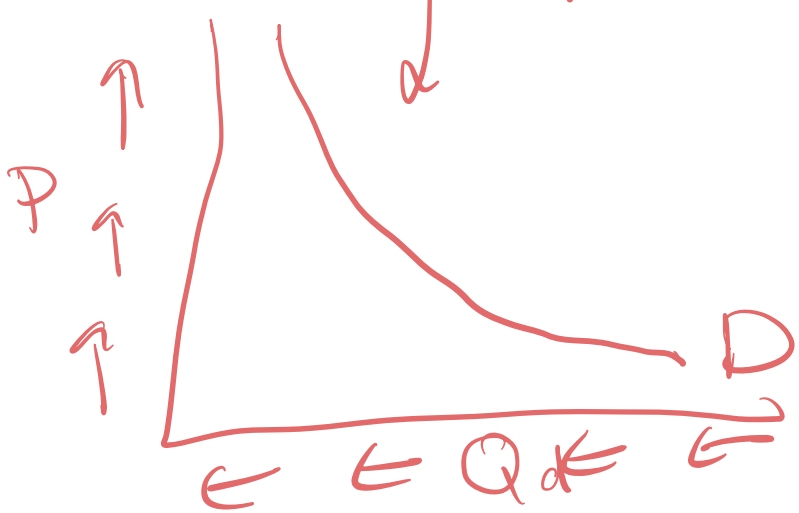
The lowest price at which someone is willing to sell an additional unit rises.

This lowest price is *marginal cost*.



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Supply



A Change in Supply

When some influence on selling plans other than the price of the good changes, there is a **change in supply** of that good.

The quantity of the good that producers plan to sell changes at each and every price, so there is a new supply curve.

When supply *increases*, the supply curve shifts *rightward*.

When supply *decreases*, the supply curve shifts *leftward*.

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★ distinguish between six factors in supply vs. demand

Supply

The six main factors that change supply of a good are

- The prices of factors of production
- The prices of related goods produced → check prices of competitors
- Expected future prices → as a seller I'd expect to sell at low/high price
  e.g. prices for winter abaya at winter season
- The number of suppliers → more sellers → more supply in the market (vice versa)
- Technology → invest in high technology to cut down cost
- State of nature
  ↳ weather, condition, floods, natural disaster, wars

e.g. damaged crops minimizes supply

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Supply

Prices of Factors of Production

If the price of a factor of production used to produce a good rises, the minimum price that a supplier is willing to accept for producing each quantity of that good rises.

So a rise in the price of a factor of production decreases supply and shifts the supply curve leftward.

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Supply

Prices of Related Goods Produced

A *substitute in production* for a good is another good that can be produced using the same resources.

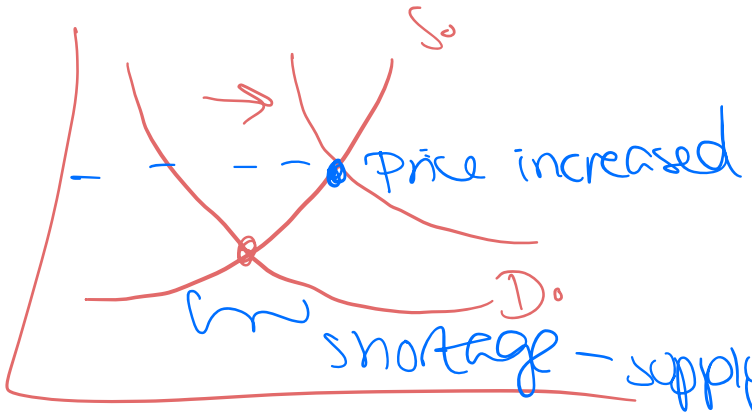
The supply of a good increases if the price of a substitute in production falls. → resources shift away from substitutes with lower prices as they contribute to lower profit, reducing the incentive

Goods are *complements in production* if they must be produced together.

The supply of a good increases if the price of a complement in production rises.

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Supply



supply increases since price increased due to demand rate

Expected Future Prices

If the price of a good is expected to rise in the future, supply of the good today decreases and the supply curve shifts leftward.

shifts sub leftward

The Number of Suppliers

The larger the number of suppliers of a good, the greater is the supply of the good. An increase in the number of suppliers shifts the supply curve rightward.

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Supply

Technology

Advances in technology create new products and lower the cost of producing existing products.

So advances in technology increase supply and shift the supply curve rightward.

The State of Nature

The state of nature includes all the natural forces that influence production—for example, the weather.

A natural disaster decreases supply and shifts the supply curve leftward.

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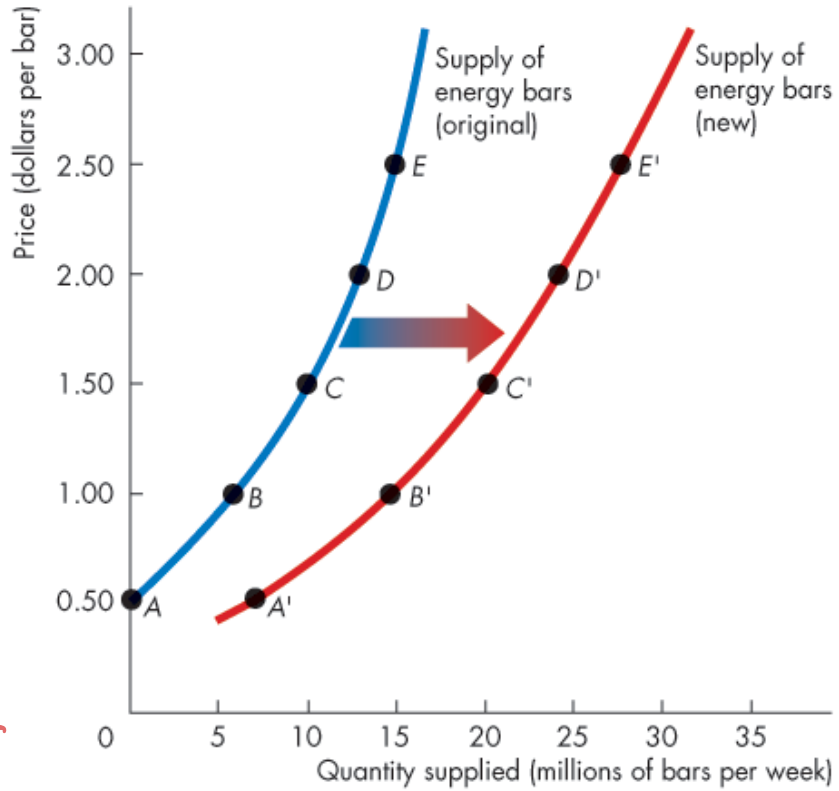
any increase in cost of production negatively affects the supplier resulting in a decrease in supply
→ thus the curve will shift leftward

Supply

Figure 3.5 shows an increase in supply.

An advance in the technology increases the supply of energy bars and shifts the supply curve rightward.

	Original supply schedule Old technology			New supply schedule New technology	
	Price (dollars per bar)	Quantity supplied (millions of bars per week)		Price (dollars per bar)	Quantity supplied (millions of bars per week)
A	0.50	0	A'	0.50	7
B	1.00	6	B'	1.00	15
C	1.50	10	C'	1.50	20
D	2.00	13	D'	2.00	25
E	2.50	15	E'	2.50	27



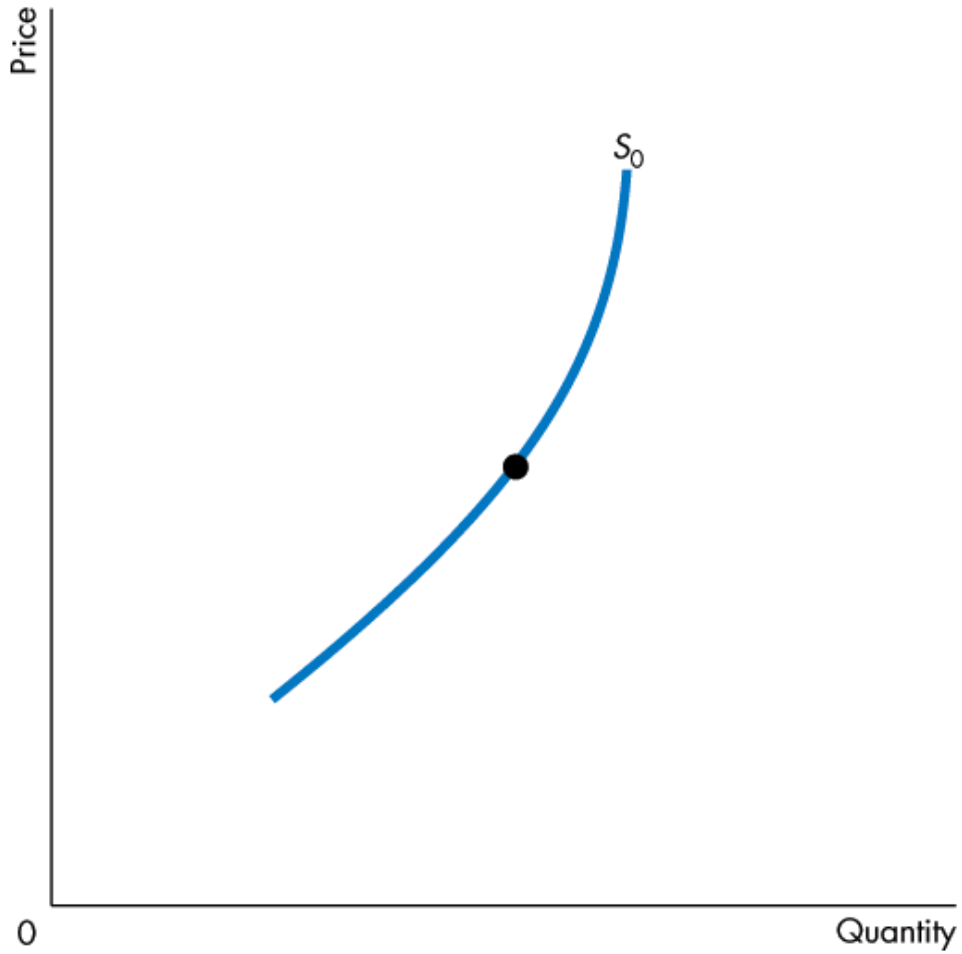
(supply increases) - right shift

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Supply

A Change in the Quantity Supplied Versus a Change in Supply

Figure 3.6 illustrates the distinction between a change in supply and a change in the quantity supplied.

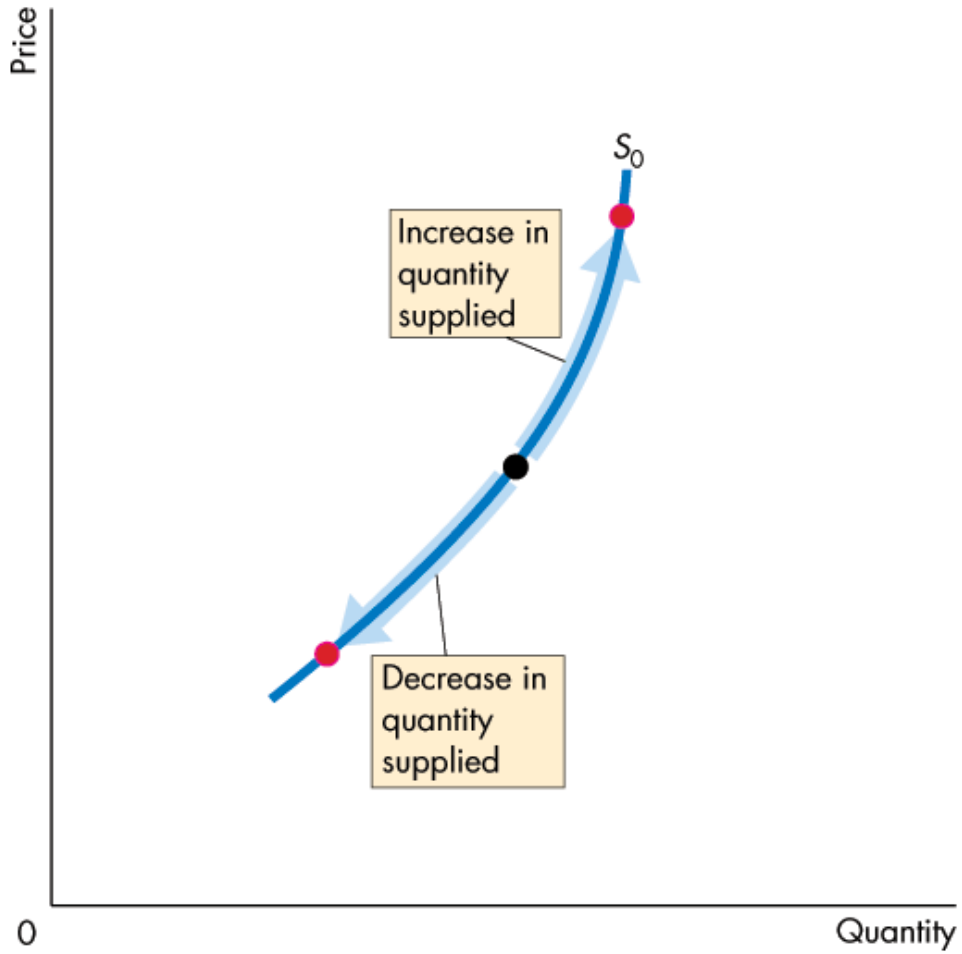


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Supply

Movement Along the Supply Curve

When the price of the good changes and other influences on sellers' plans remain the same, the quantity supplied changes and there is a movement along the supply curve.



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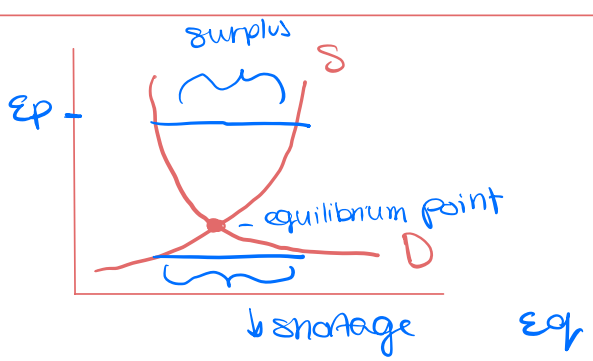
market clearance - where both entities agree



→ both entities meet at the market
→ both interact
→ determine price that they can agree on

$Q_d = Q_s$ whenever an equilibrium position is encountered

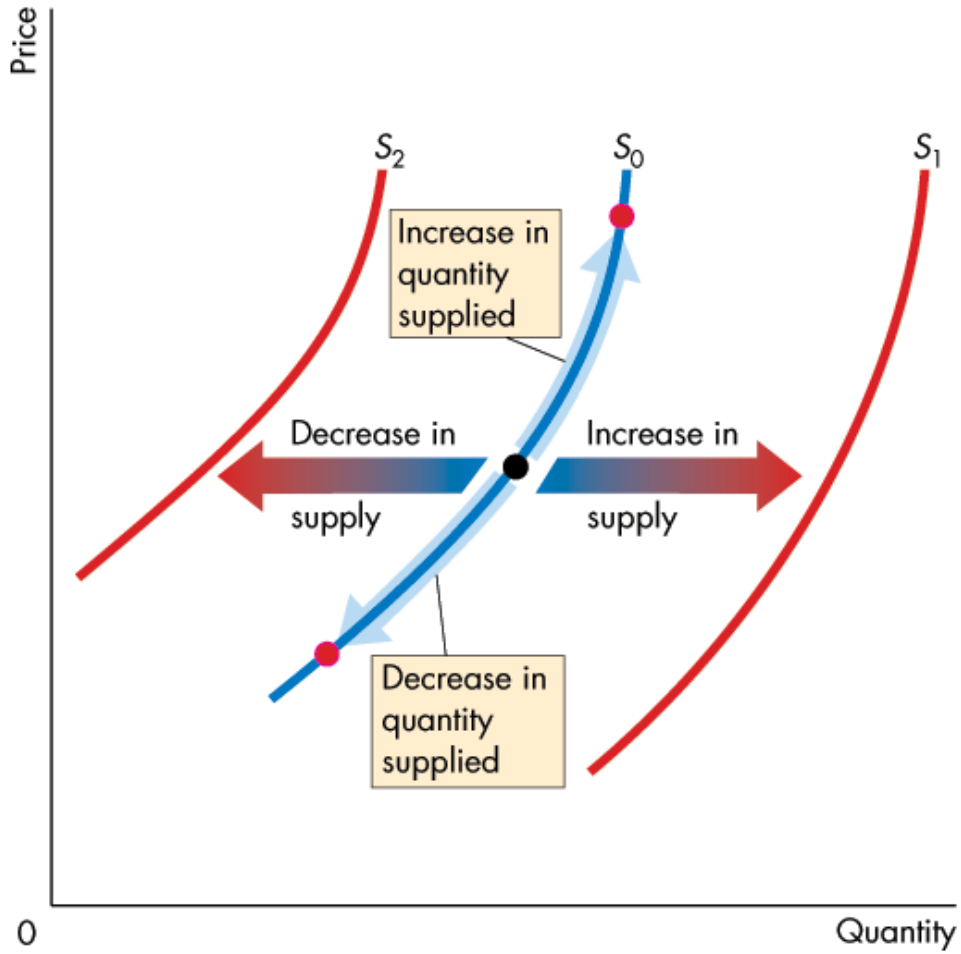




Supply

A Shift of the Supply Curve

If the price remains the same but some other influence on sellers' plans changes, supply changes and the supply curve shifts.



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Market Equilibrium

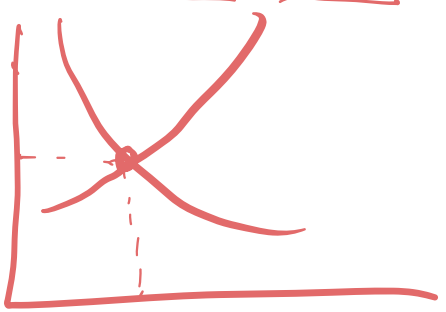
Equilibrium is a situation in which opposing forces balance each other.

A market equilibrium occurs when the price balances the plans of buyers and sellers.

The **equilibrium price** is the price at which the quantity demanded equals the quantity supplied.

The **equilibrium quantity** is the quantity bought and sold at the equilibrium price.

- Price regulates buying and selling plans.
- Price adjusts when plans don't match.



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If Qd > Qs = shortage (less stock, more demand)
If Qd < Qs = surplus (more stock, less demand)

Market Equilibrium

schedule used to determine equilibrium

Price (dollars per bar)	Quantity demanded (millions of bars per week)	Quantity supplied (millions of bars per week)
0.50	22	0
1.00	15	6
1.50	**10**	**10**
2.00	7	13
2.50	5	15

→ shortage

→ surplus

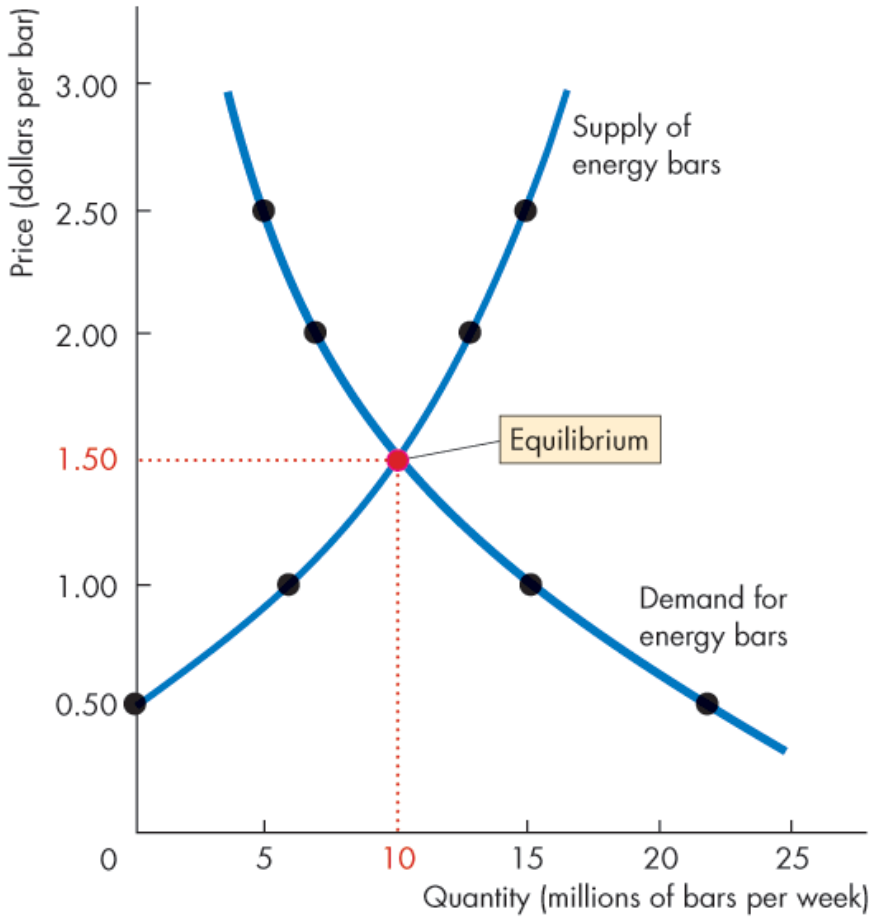


Figure 3.7 illustrates the market equilibrium—the price at which quantity demanded equals quantity supplied.

Qs & Qd intersect at a given price

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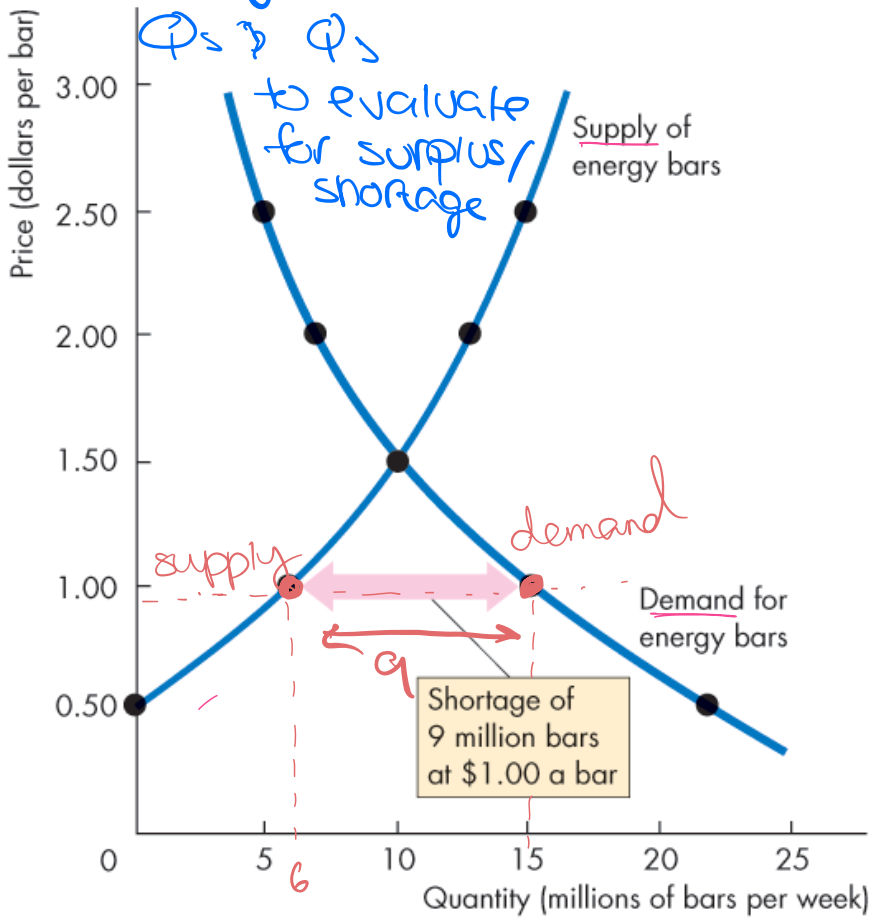
Market Equilibrium

Price as a Regulator

Price (dollars per bar)	Quantity demanded	Quantity supplied	Shortage (−) or surplus (+)
	(millions of bars per week)		
0.50	22	0	−22
1.00	15	6	−9
1.50	10	10	0
2.00	7	13	+6
2.50	5	15	+10

If the price is $1.00 a bar, the quantity demanded *exceeds* the quantity supplied.

There's a *shortage* of 9 million energy bars.



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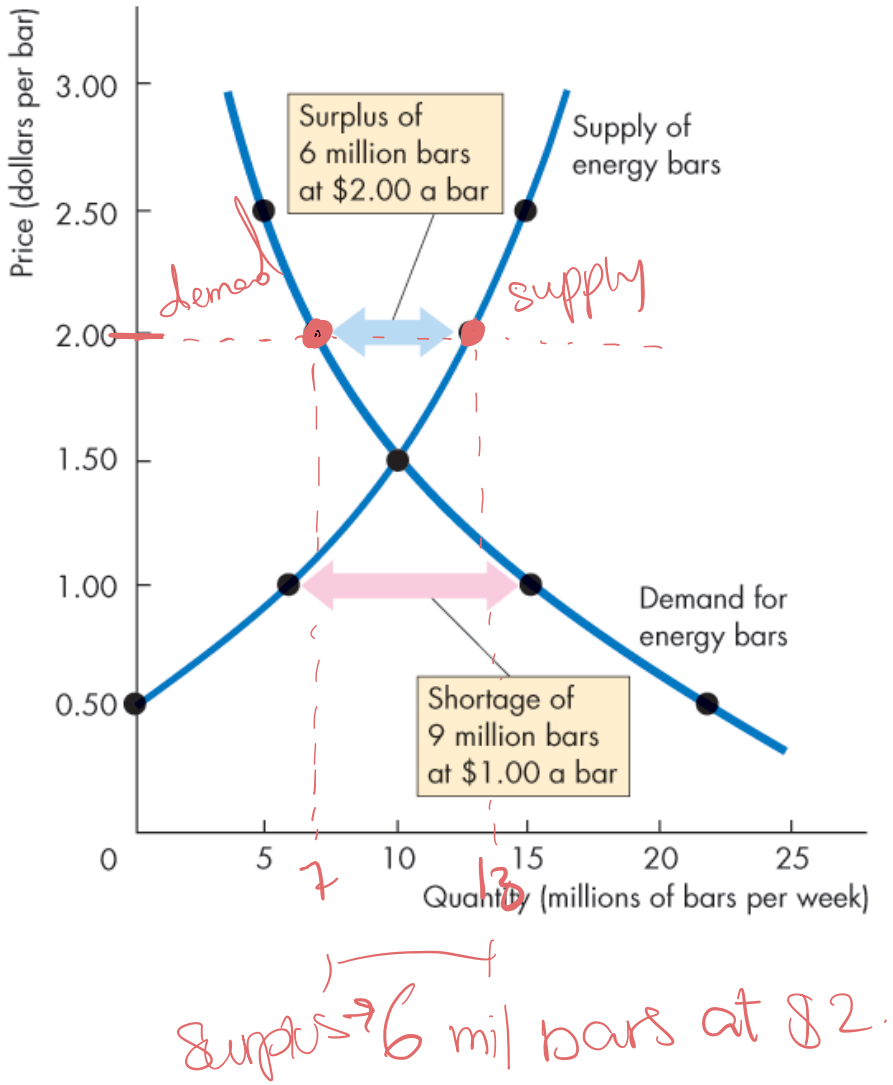
Market Equilibrium

Price as a Regulator

Price (dollars per bar)	Quantity demanded	Quantity supplied	Shortage (−) or surplus (+)
	(millions of bars per week)		
0.50	22	0	−22
1.00	15	6	−9
1.50	10	10	0
2.00	7	13	+6
2.50	5	15	+10

If the price is $2.00 a bar, the quantity supplied *exceeds* the quantity demanded.

There's a *surplus* of 6 million energy bars.



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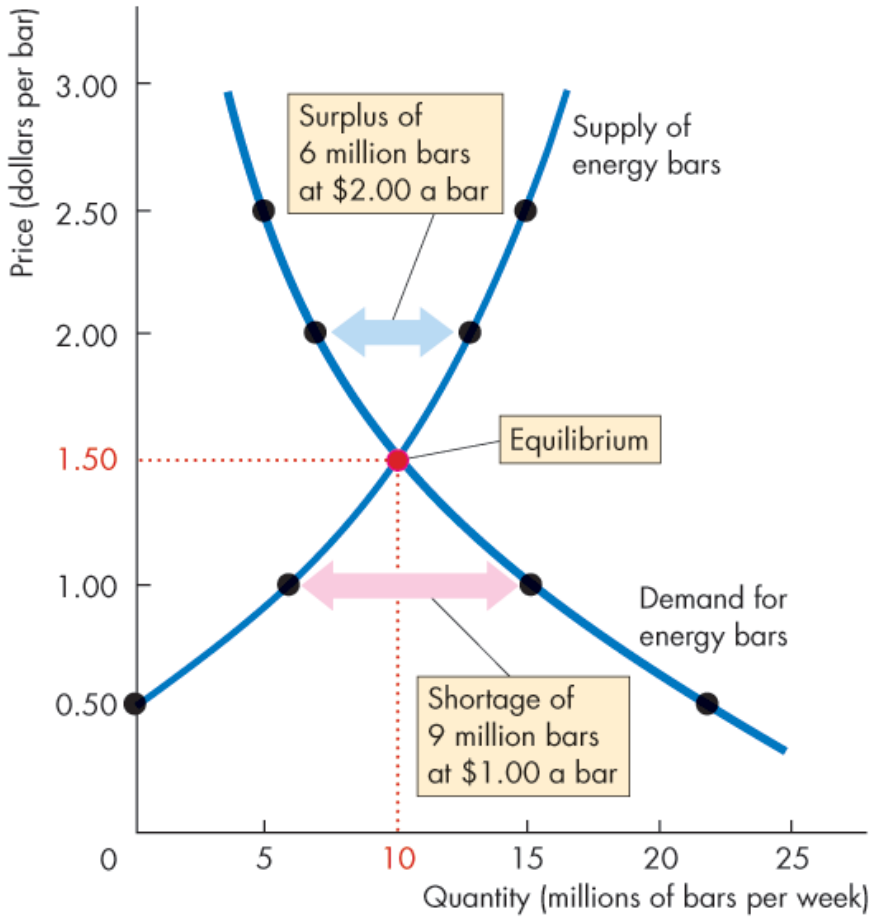
Market Equilibrium

Price as a Regulator

Price (dollars per bar)	Quantity demanded	Quantity supplied	Shortage (−) or surplus (+)
	(millions of bars per week)		
0.50	22	0	−22
1.00	15	6	−9
1.50	**10**	**10**	**0**
2.00	7	13	+6
2.50	5	15	+10

If the price is $1.50 a bar, the quantity supplied *equals* the quantity demanded.

No shortage or surplus of bars.



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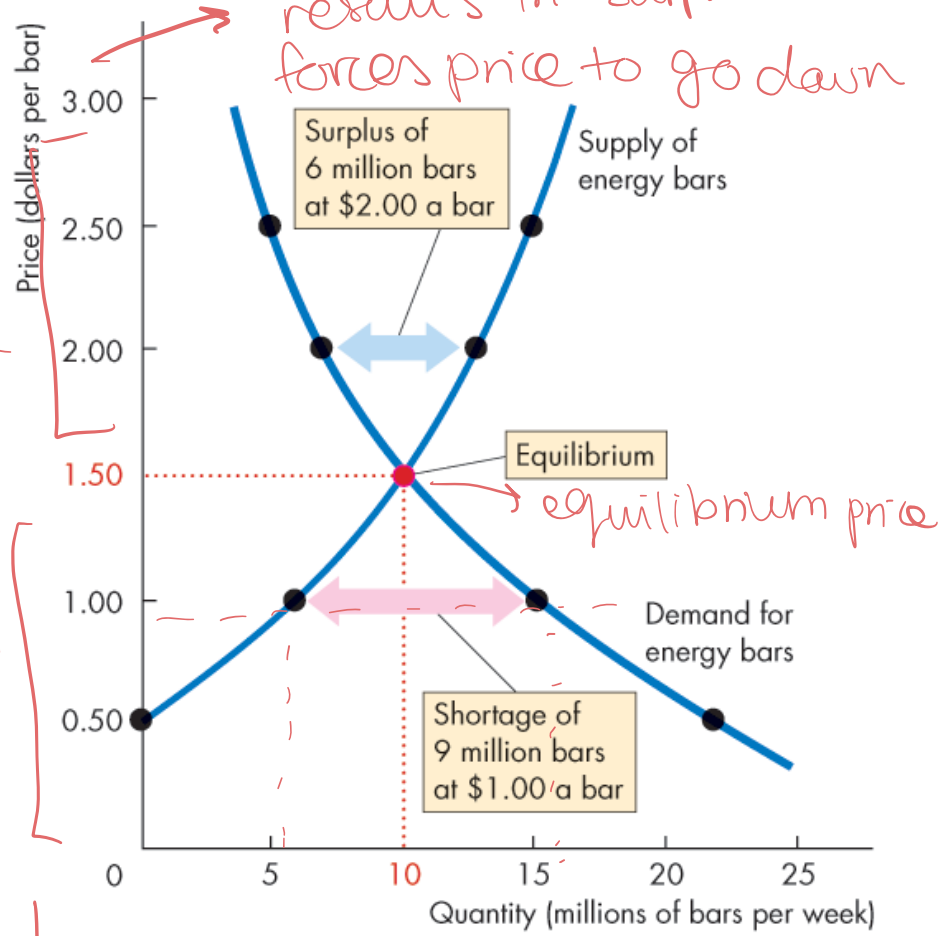
Market Equilibrium

Price Adjustments

At prices below the equilibrium price, a *shortage* forces the price up.

At prices above the equilibrium price, a *surplus* forces the price down.

At the equilibrium price, buyers' plans and sellers' plans agree and the price doesn't change until an event changes demand or supply.



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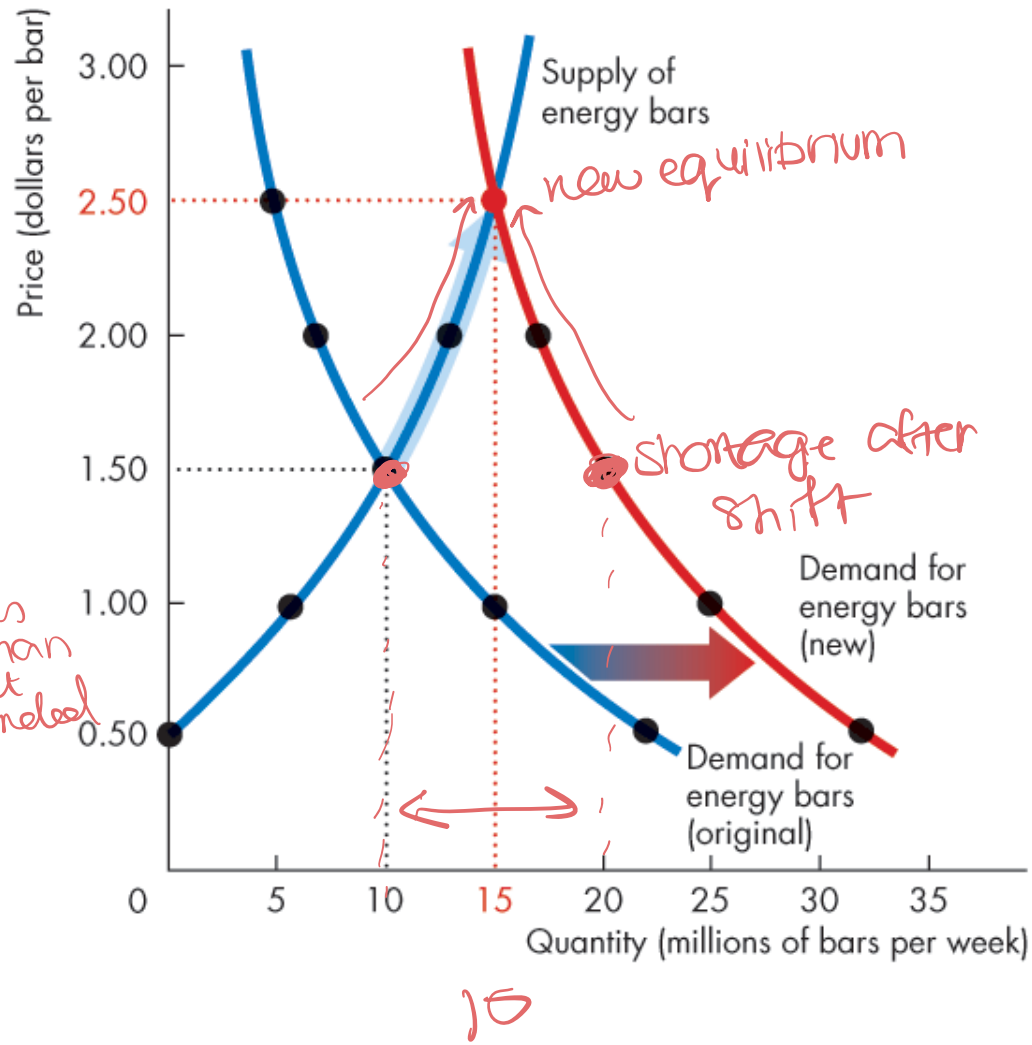
Predicting Changes in Price and Quantity

An Increase in Demand

Figure 3.8 shows that when demand increases the demand curve shifts rightward.

At the original price, there is now a *shortage*.

The price rises, and the quantity supplied increases along the supply curve.



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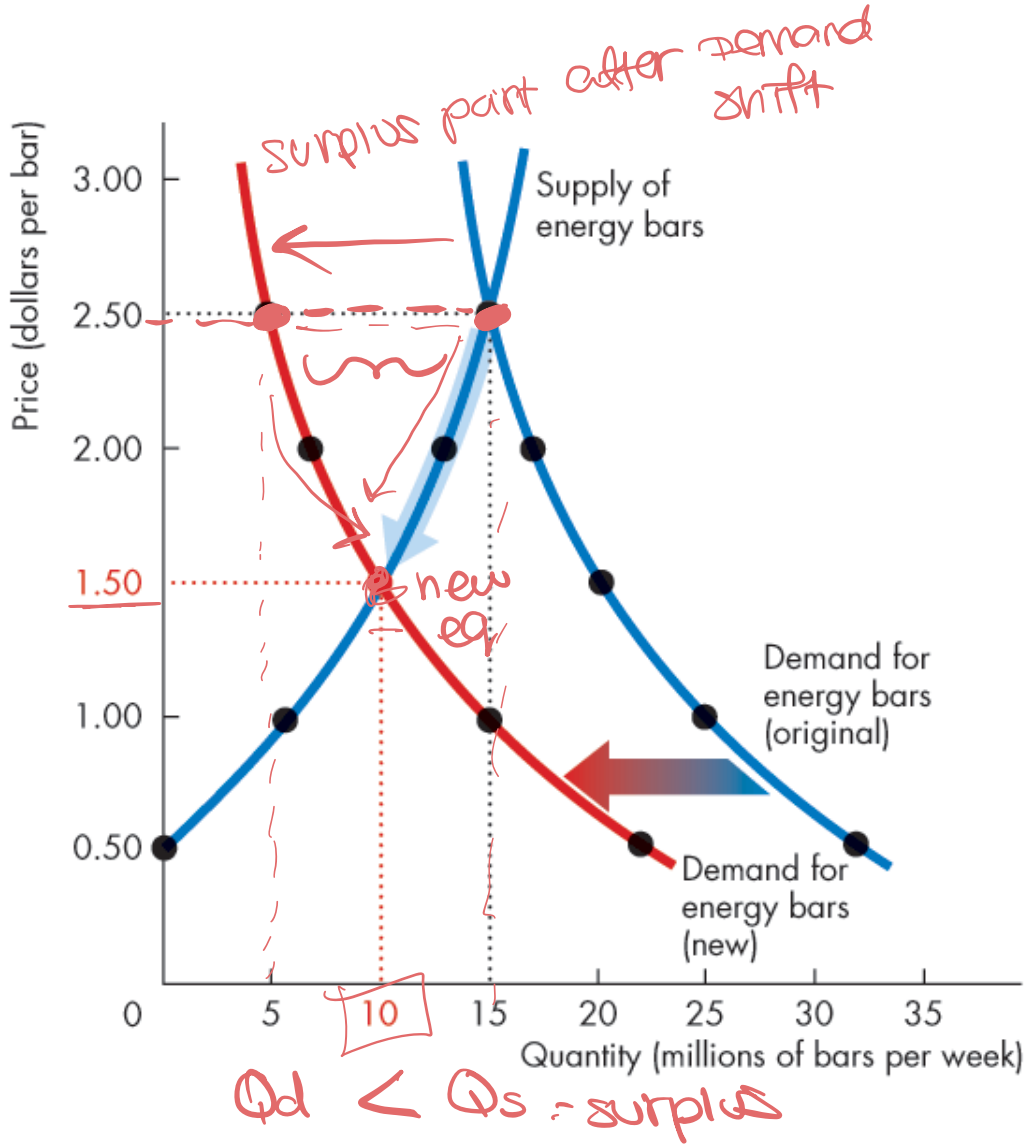
Predicting Changes in Price and Quantity

A Decrease in Demand

The figure shows that when demand decreases the demand curve shifts leftward.

At the original price, there is now a *surplus*.

The price falls, and the quantity supplied decreases along the supply curve.



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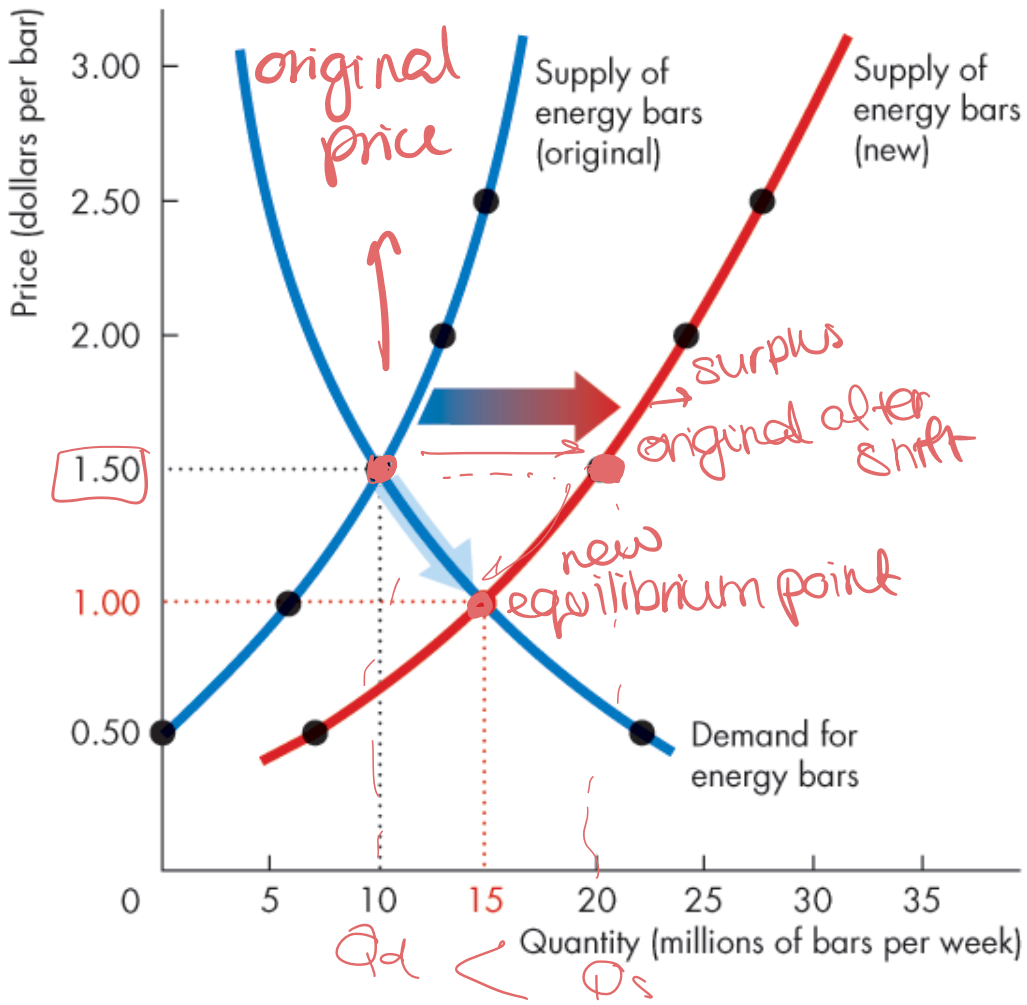
Predicting Changes in Price and Quantity

An Increase in Supply

Figure 3.9 shows that when supply increases the supply curve shifts rightward.

At the original price, there is now a *surplus*.

The price falls, and the quantity demanded increases along the demand curve.



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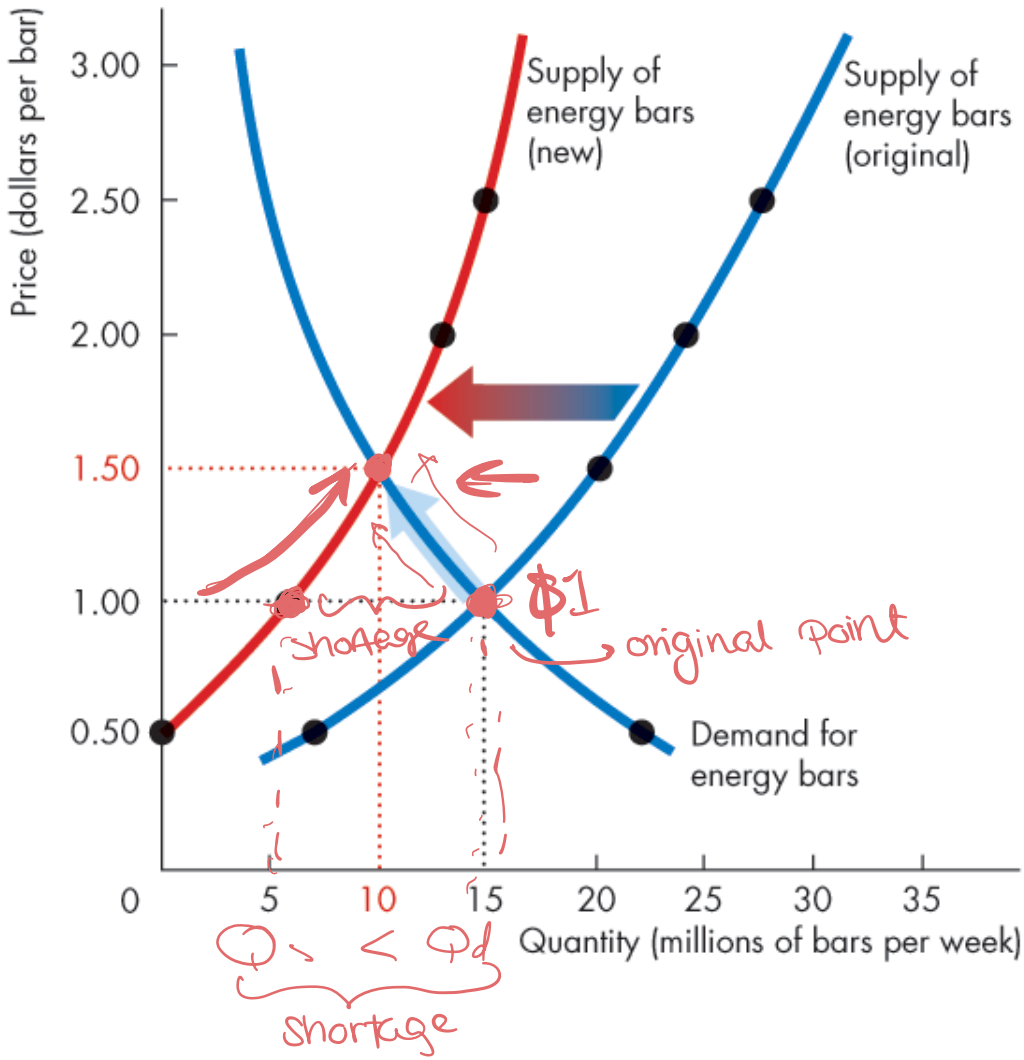
Predicting Changes in Price and Quantity

A Decrease in Supply

The figure shows that when supply decreases the supply curve shifts leftward.

At the original price, there is now a *shortage*.

The price rises, and the quantity demanded decreases along the demand curve.



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Predicting Changes in Price and Quantity

Changes in Both Demand and Supply

A change in both demand and supply changes the equilibrium price and the equilibrium quantity.

Figure 3.10 illustrates changes in the same direction.

Figure 3.11 illustrates changes in opposite directions.

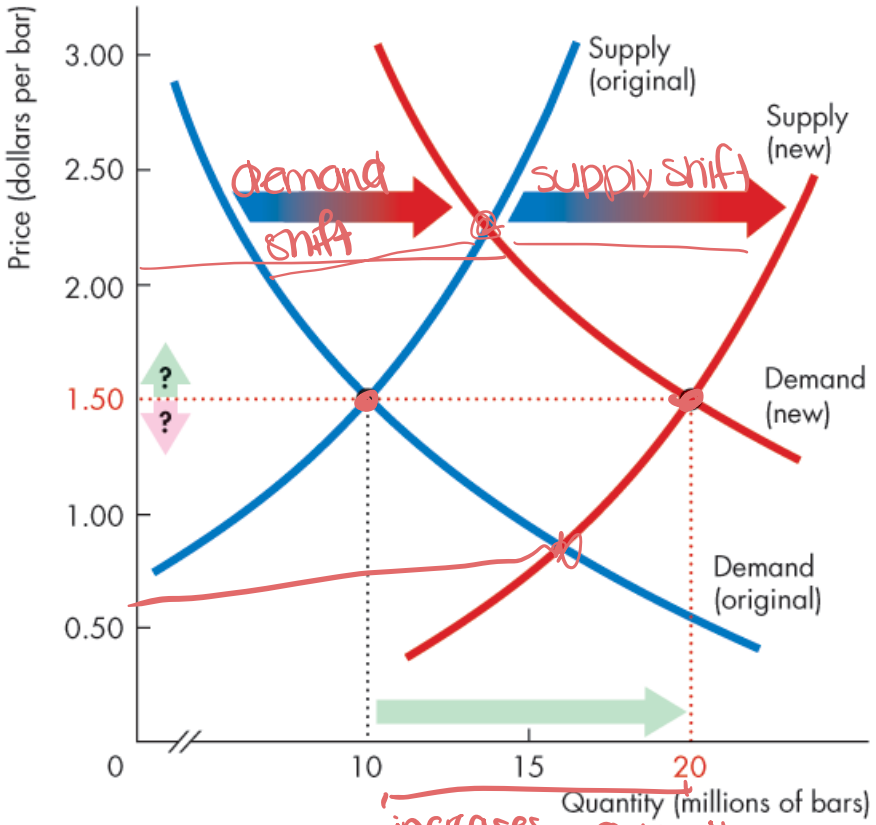
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Predicting Changes in Price and Quantity

Demand and Supply Change in the Same Direction

An increase in demand and an increase in supply *increase* the equilibrium quantity.

The change in equilibrium price is *uncertain* because the increase in demand raises the price and the increase in supply lowers it.



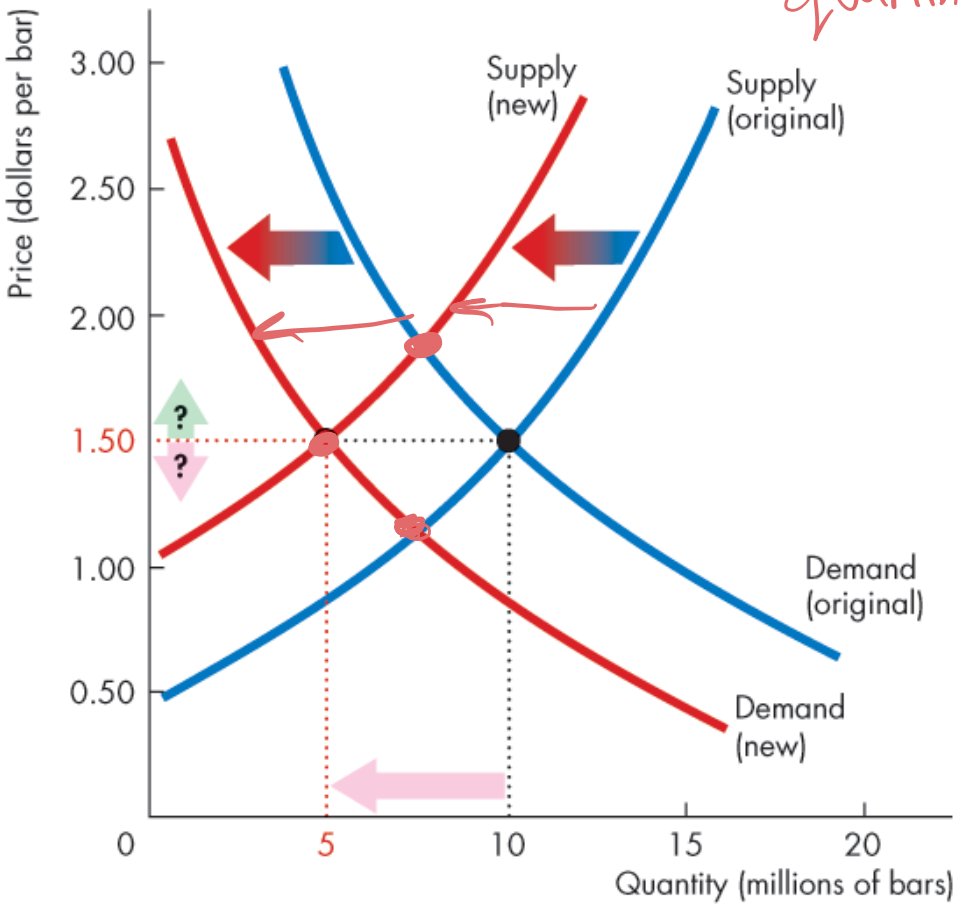
(a) Increase in both demand and supply

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Predicting Changes in Price and Quantity

A decrease in both demand and supply *decreases* the equilibrium quantity.

The change in equilibrium price is *uncertain* because the decrease in demand lowers the price and the decrease in supply raises the price.



(b) Decrease in both demand and supply

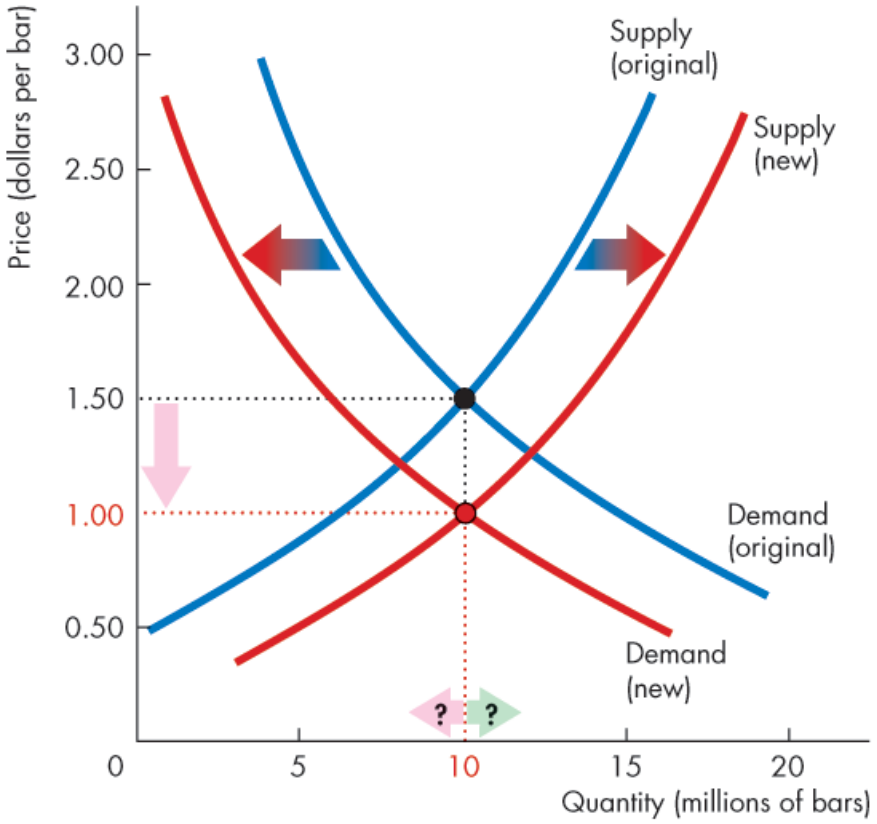
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Predicting Changes in Price and Quantity

Demand and Supply Change in Opposite Directions

An increase in supply and a decrease in demand *lowers* the equilibrium price.

The change in equilibrium quantity is *uncertain* because the decrease in demand decreases the quantity and the increase in supply increases it.



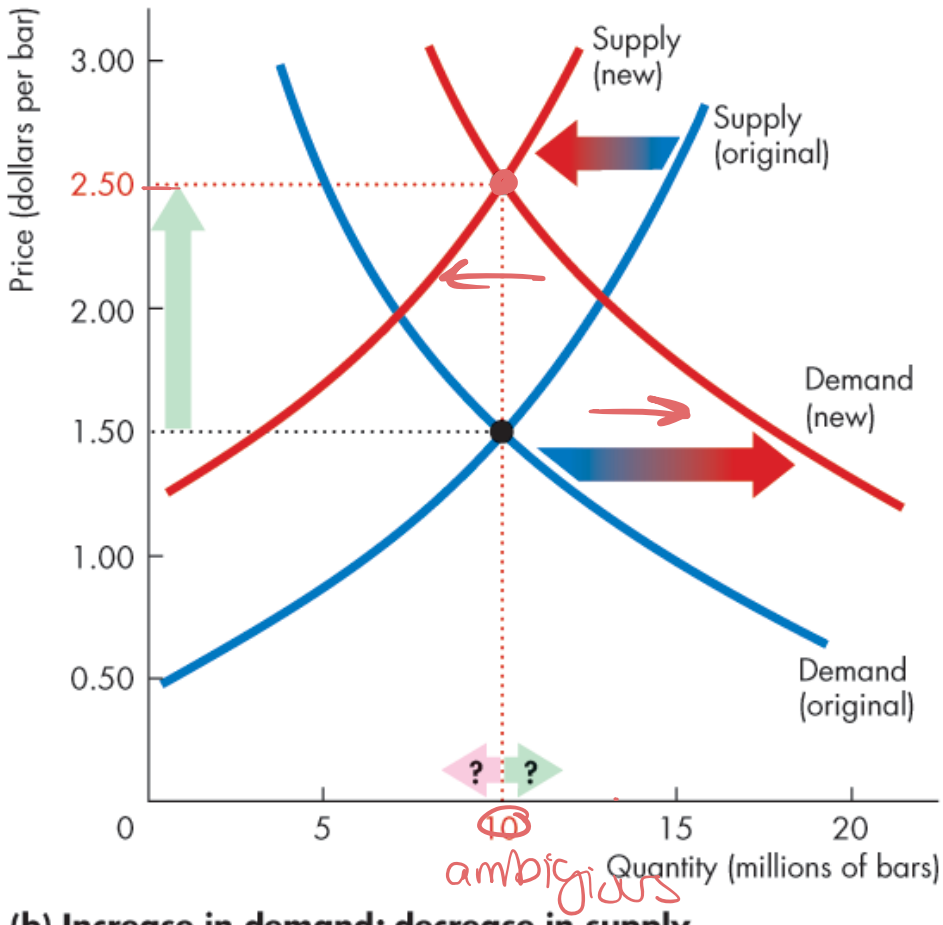
(a) Decrease in demand; increase in supply

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Predicting Changes in Price and Quantity

A decrease in supply and an increase in demand and *raises* the equilibrium price.

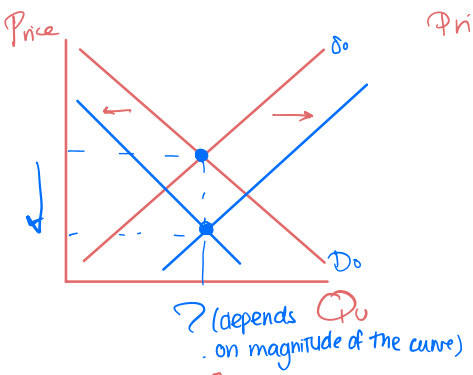
The change in equilibrium quantity is *uncertain* because the decrease in supply decreases the quantity and the increase in demand increases it.



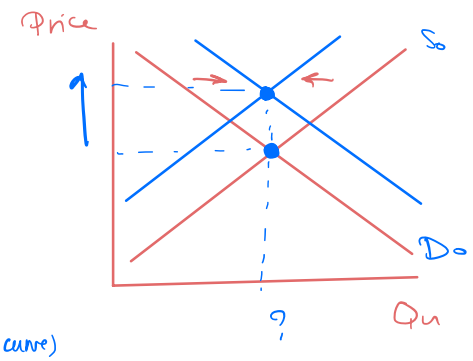
(b) Increase in demand; decrease in supply

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Change in Supply	Change in Demand	Effect on Equilibrium Price	Effect on Equilibrium Quantity
1. Increase	Decrease	Decrease	Indeterminate
2. Decrease	Increase	Increase	Indeterminate
3. Increase	Increase	Indeterminate	Increase
4. Decrease	Decrease	Indeterminate	Decrease

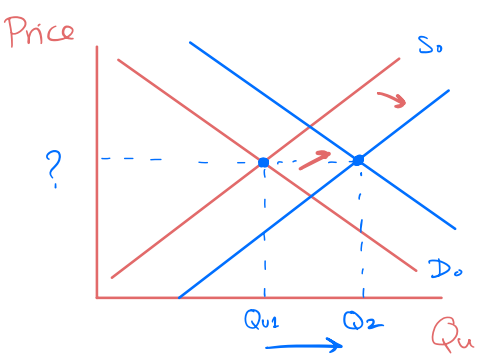


ΔD↓ ΔS↑
- eq price decreases
- eq quantity is determined by magnitude of the curves

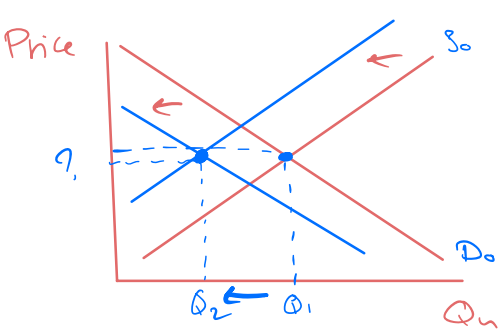


ΔD↑ ΔS↓
- eq price increases
- eq quantity is determined by magnitude of the curves

cases when both forces oppose each other



ΔD↑ ΔS↑
- Equilibrium quantity increases
- Equilibrium Price is indeterminant



Δ↓ ΔS↓
- Equilibrium quantity decreases
- Equilibrium price is ambiguous